Nagar YuwakShikshanSanstha's

YeshwantraoChavan College of Engineering

(An Autonomous Institution affiliated to RashtrasantTukadojiMaharaj Nagpur University)

(Accredited 'A' Grade by NAAC)

Hingna Road, Wanadongri, Nagpur - 441 110

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Form No YI-04

Department of Electronics Engineering

STUDENT'S DIARY/ LOG

Your Name: Khushi Narendra Atara

Your Institute: Yeshwantrao Chavan College of Engineering Nagpur

Company where internships are carried out : Cognizant Technology Solutions Pvt. Ltd

Period: From: 31/01/25 To: 13/05/26

Week: 3		DATE: From	16/02/26 to 20/02/26
Dept./Division	Programmer Analyst Trainee	Name of Project	Training and Internship Program
Name of HOD/ Supervisor With e-mail id		Name: Ashly M. Benny	
		Email Id: ashly.bennym@cognizant.com	
Main points of the week		Core Java – Stage 1 (Days 6–10): Collections, Exception Handling & Java 8 Features	

- Studied Exception Handling: try-catch block, multiple catch, finally, throw and throws keywords.
- Learned Java Collections Framework: ArrayList, LinkedList, HashSet, HashMap, TreeSet, TreeMap.
- Understood Set Interface (HashSet, LinkedHashSet, TreeSet) and Map Interface (HashMap, LinkedHashMap).
- Studied Iterator and ListIterator for traversing collections.
- Learned Java 8 Features: Lambda Expressions, Stream API (filter, map, reduce), and Optional class.
- Understood Generics: generic classes, methods, and wildcards; Annotations (@Override, @Deprecated).
- Studied Design Patterns: Singleton, Factory, Observer, and Strategy patterns.
- Completed mandatory Tekstac hands-on: WordZee, Fishing Competition, Brilliant Restaurant, Investment Calculation.
- Practiced Eclipse exercises on Arrays, String handling, Collections, and Exception handling.
- Attended Business Mentor Connect session and Practice ICT Assessment on Day 7 (17/02/26).

Signature and stamp of Industry Supervisor

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Fortnightly Progress Report

(To be mailed every alternate Monday to Project Guide, designated person(s) of the company, your HoD and the concerned Dean from your institute)

1. Company Overview:

Cognizant Technology Solutions is a leading global IT services and consulting company headquartered in the USA, with a major delivery presence across India. It specializes in digital transformation, cloud computing, AI, automation, software engineering, and quality assurance. Cognizant's QEA (Quality Engineering and Assurance) division focuses on intelligent and automated quality assurance, modernisation assurance, and experience assurance for enterprise clients across healthcare, finance, and technology sectors.

2. Who did you interact with in this period? Which departments did you visit/work in?

During Week 3, I interacted with Cognizant Academy trainers, fellow GenC trainees, and a BU Mentor during the Business Mentor Connect session held on Day 7 (17/02/26). All interactions took place within the Training division. The trainer-led Master Sessions covered Core Java advanced topics for Days 6 to 10.

3. List down (maximum) three activities carried out by you in this period. Describe each one briefly.

Activity 1 – Java Collections & Exception Handling (ILT): Attended Master Sessions covering the Java Collections Framework (ArrayList, HashMap, HashSet, TreeMap) and Exception Handling (try-catch, finally, custom exceptions). Practiced all examples in Eclipse.

Activity 2 – Tekstac Hands-On Exercises: Completed mandatory hands-on problems – WordZee, Fishing Competition, Brilliant Restaurant, and Investment Calculation – on the Tekstac platform with auto-evaluation.

Activity 3 – Business Mentor Connect & ICT Practice: Participated in the BU Mentor Connect session on Day 7 for industry insights. Also attempted the Practice ICT Assessment on the same day to gauge understanding of Java concepts covered so far.

4. What problems did you face and what did you do to address them? What was the learning that resulted from each of them?

Found it challenging to understand the differences between HashMap, LinkedHashMap, and TreeMap in terms of ordering and performance. Clarified doubts during the Master Session and referred to the Udemy course and GeeksForGeeks resources. This helped in understanding when to use each collection type based on the use case.

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5. What process(s)/system(s) did you familiarize yourself with/study as part of your work or out of your interest?

Practiced Java Collections and Exception Handling on the Tekstac platform. Additionally, explored GeeksForGeeks resources on Serial and Parallel Sorts in Java and Java 8 Stream API as recommended in the learning guide. Also referred to the Udemy course 'Java In-Depth: Become a Complete Java Engineer' (Sections 16 and 17) for Collections Framework and Generics.

6. If you have seen anything where an improvement is possible/required, please mention/describe that here.

Including more real-world use-case examples while explaining Design Patterns (Singleton, Factory) during Master Sessions would help trainees better understand when and why to use each pattern in enterprise software development.

7. What did you find novel/exciting/valuable during this period?

Learning Java 8 features such as Lambda Expressions and the Stream API was the most exciting part of this week. Understanding how streams allow functional-style operations on collections made code much cleaner and more readable. The Business Mentor Connect session also provided valuable industry perspective on how these concepts are applied in real projects.

8. What according to you, can be of use to your institute and students and in what ways?

Teaching Java Collections Framework and Java 8 features (Streams, Lambda) as part of the academic Java curriculum would greatly benefit students. These are industry-standard topics that are heavily used in enterprise applications, and early exposure would reduce the learning curve when students join corporate training programs.

9. Have you been able to complete everything that you had planned last fortnight? If not, what have you not completed and why?

Yes, all planned activities for Week 3 were completed successfully. The mandatory Tekstac hands-on exercises, Eclipse practice programs, Udemy course sections, and the Practice ICT Assessment were all finished within the scheduled time.

Khushi N. Atara

Name and Signature of Student.

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